

Customer Sales Analysis

Week 5 — Advanced Data Manipulation with Pandas

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This report presents the analysis of sales and customer data, covering customer value, sales performance, regional and product analysis, retention, seasonality, cross-selling opportunities, and business recommendations.

KPI	Result
Total Revenue	■12,365,048.00
Total Orders	100
Total Quantity Sold	478
Total Customers	500
Average Order Value	■123,650.48
Retention Rate	89.40%
Churn Rate	10.60%

1. Executive Summary

The project used Python and Pandas to analyze 100 sales transactions and 500 customer records. The workflow covered data exploration, cleaning, feature engineering, grouping and aggregation, conditional filtering, string operations, datetime processing, merging, pivot tables, customer value, churn and retention, seasonal analysis, cross-selling analysis, and visualization. Total revenue was ■12,365,048.00 with an average order value of ■123,650.48. North was the leading region and Laptop was the leading product by revenue. Customer retention was 89.40%.

2. Dataset and Methodology

The sales dataset contains Date, Product, Quantity, Price, Customer_ID, Region, and Total_Sales. The customer dataset contains Customer_ID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, and Churn. The analysis created Year, Month, Month_Name and Quarter features and validated Calculated_Sales against Total_Sales.

3. Data Quality and Validation

The source datasets contained no missing values or duplicate records. Sales calculations were validated successfully. Customer identifiers initially used different formats, so they were standardized before merging. The corrected merge matched all 100 sales records to customer records, with zero unmatched records. The final merge validation passed.

Validation	Result
Sales records	100
Customer records	500
Missing values	None identified
Duplicate records	None identified
Sales calculation	PASS
Matched customer records	100
Unmatched records	0
Merge validation	PASS

4. Customer Analysis

Customer-level aggregation identified CUST016 as the top customer by sales revenue with ■373,932.00. Customer lifetime value was also segmented into Low, Medium, and High Value groups. The High Value segment had the highest average lifetime value.

Customer Segment	Customers	Average Lifetime Value
Low Value	167	■1,635.75
Medium Value	166	■4,222.65
High Value	167	■6,855.15

5. Regional and Product Performance

North was the strongest region with █3,983,635.00 in revenue, followed by South, East, and West. Laptop was the best-performing product with █3,889,210.00 in revenue.

Region	Revenue	Quantity	Orders
North	█3,983,635	147	28
South	█3,737,852	143	27
East	█2,519,639	94	19
West	█2,123,922	94	26

Product	Revenue	Quantity	Orders
Laptop	█3,889,210	136	24
Tablet	█2,884,340	127	26
Phone	█2,859,394	101	20
Headphones	█1,384,033	48	15
Monitor	█1,348,071	66	15

6. Monthly and Seasonal Analysis

Sales were grouped by month and quarter to identify trends. Q1 2024 was the strongest sales period, generating █11,261,580.00. Q2 2024 generated █1,103,468.00. Monthly and regional pivot tables were also used to compare performance.

7. Customer Retention and Churn

Out of 500 customers, 447 were retained and 53 were churned. The resulting retention rate was 89.40% and the churn rate was 10.60%. This indicates strong overall retention while still leaving an opportunity for targeted churn-reduction campaigns.

8. Cross-Selling Analysis

No customers in the available sales transactions purchased more than one different product. Therefore, no reliable product-pair pattern could be established from this dataset. This should be treated as a limitation of the available transaction history, not as evidence that cross-selling is ineffective.

9. Sales Performance Dashboard

A five-visualization dashboard was created covering the monthly sales trend, sales by region, sales by product, top customers, and customer retention versus churn. The dashboard provides a concise visual summary of the main analytical findings.

10. Key Business Insights

- Total revenue: ■12,365,048.00 across 100 orders.
- Top customer: CUST016 with ■373,932.00 revenue.
- Top region: North with ■3,983,635.00 revenue.
- Best-selling product: Laptop with ■3,889,210.00 revenue.
- Retention rate: 89.40%; churn rate: 10.60%.
- Strongest quarter: Q1 2024 with ■11,261,580.00 revenue.
- No multi-product customer combinations were available for cross-selling analysis.

11. Business Recommendations

- Focus retention campaigns on high-value customers and customers at risk of churn.
- Use North-region performance as a benchmark and investigate lower-performing regions.
- Prioritize successful products such as Laptop while improving weaker categories.
- Provide personalized offers and loyalty benefits to high-value customers.
- Plan inventory and promotions around strong seasonal periods.
- Collect broader multi-product purchase history to enable stronger cross-selling analysis.

12. Conclusion

The Customer Sales Analysis project demonstrates practical use of advanced Pandas techniques to transform raw sales and customer information into business insights. The project successfully implemented grouping and aggregation, multiple-condition filtering, string operations, datetime processing, merging, pivot tables, customer analysis, retention analysis, seasonal analysis, and dashboard visualization. Final validation confirmed correct sales calculations and a successful customer-sales merge with 100 matched records and zero unmatched records.